divide (A , B)

-- inputs: A is an integer and B ≠ 0

-- outputs : q and r such that A = q X B + r and

-- 0 < r < B

q, r : = 0, A

while r ≥ B

q, r := q + 1, r - B

The algorithm is presented as a table in Figure 7.1.

We can execute the algorithm step-by-step for a test input, say, (A, B) = (22, 5). Each row of Table 7.1 shows one iteration — the evaluation of the expressions and the values of the variables at the end of an iteration. Note that the evaluation of the expression uses the values of the variables from the previous row. Output variables q and r change their values in each iteration. Input variables A and B do not change their values. Iteration 0 shows the values just before the loop starts. At the end of iteration 4, condition (r ≥ B) = (2 ≥ 5) is false, and hence the loop ends with (q, r) = (4, 2).

iteration	q	q+1	r	r-B	A	B
0	0		22		22	5
1	1	0+1	17	22-5		
2	2	1 + 1	12	17-5		
3	3	2+1	7	12-5		
4	4	3+1	2	7-5		

Table 7.1: Step by step execution of divide (22, 5)

Example 7.5. In the Chameleons of Chromeland problem of Example 1.3, suppose two types of chameleons are equal in number. Construct an algorithm that arranges meetings between these two types so that they change their color to the third type. In the end, all should display the same color.

Let us represent the number of chameleons of each type by variables a, b and c and their initial values by A, B and C, respectively. Let a = b be the input property. The input-output relation is a = b = 0 and c = A+B+C. Let us name the algorithm monochromatize. The algorithm can be specified as

monochromatize(a, b, c)

-- inputs: a=A, b=B, c=C, a=b

-- outputs : a = b = 0 , c = A+B+C

In each iterative step, two chameleons of the two types (equal in number) meet and change their colors to the third one. For example, if A, B, C = 4, 4, 6, then the series of meetings will result in

iteration	a	b	c
0	4	4	6
1	3	3	8
2	2	2	10
3	1	1	12
4	0	0	14

Table 7.2: Series of meetings between two types of chameleons equal in number.

In each meeting, a and b each decreases by 1, and c increases by 2. The solution can be expressed as an iterative algorithm.

monochromatize(a, b, c)

-- inputs: a=A, b=B, c=C, a=b

-- outputs: a = b = 0, c = A+B+C

while a > 0

a, b, c := a-1, b-1, c+2

The algorithm is depicted in the flowchart of Figure 7.7.

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